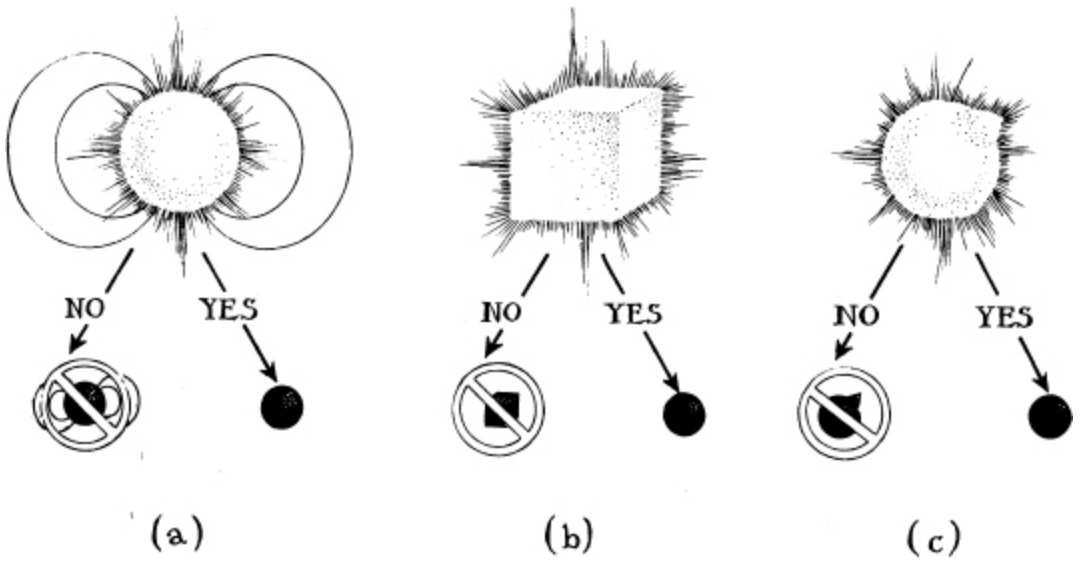
conjecture" is correct. [Left: courtesy Vitaly Ginzburg; right: courtesy Werner Israel.]

To test this speculation by computing the full details of the star's implosion would have been exceedingly difficult, so Ginzburg did the second best thing. Like Oppenheimer in his first crude exploration of what happens when a star implodes (Chapter 6), Ginzburg examined a sequence of static stars, each one more compact than the previous one, and all with the same number of magnetic field lines threading through their interiors. This sequence of static stars should mimic a single imploding star, Ginzburg reasoned. Ginzburg derived a formula that described the shapes of the magnetic field lines for each of the stars in his sequence—and found a great surprise. When a star was nearly at its critical circumference and beginning to form a black hole around itself, its gravity sucked its magnetic field lines down onto its surface, plastering them there tightly. When the black hole was formed, the plastered-down field lines were all inside its horizon. No field lines remained, sticking out of the hole (Figure 7.3a). This did not bode well for Ginzburg's idea of how to power quasars, but it did suggest an intriguing possibility: When a magnetized star implodes to form a black hole, the hole might well be born with no magnetic field whatsoever.



7.3 Some examples of the "no-hair conjecture": (a) When a magnetized star implodes, the hole it forms has no magnetic field. (b) When a square star implodes, the hole it forms is round, not square. (c) When a star with a mountain on its surface implodes, the hole it forms has no mountain.